

Validation Evidence Applicability Module (VEAM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation Evidence Fitness Standard (VEFS)

Operational Layer: Validation Evidence Fitness Governance Layer

Category: AI & Interpretation

Subcategory: Validation Governance

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Governed Space: Validation Evidence Applicability

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Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation Evidence Applicability

Canonical Definition

Validation Evidence Applicability Module (VEAM) defines the governance framework for determining whether Relevant Validation Evidence generated, observed, measured, recorded, or established under specific conditions can legitimately be applied to the actual Validation Object, version, configuration, population, environment, jurisdiction, operational state, temporal condition, and Validation Scope for which validation is being performed.

It establishes the conditions under which evidence relevance becomes

Evidence Applicability within a defined validation context.

Operational Role

VEAM serves as the second evidence-qualification gate within the Validation Evidence Fitness Standard.

The preceding Validation Evidence Relevance Module asks:

Does this evidence meaningfully address the validation question?

VEAM asks:

Can this relevant evidence legitimately be applied to this particular Validation Object and the conditions for which the validation conclusion is intended?

The governed progression is:

Relevant Validation Evidence → Applicability Dimensions → Condition Comparison → Transfer Assessment → Applicability Determination → Applicable Validation Evidence

VEAM prevents relevant evidence from being generalized beyond the conditions it can legitimately support.

Module Operational Space

VEAM governs:

- evidence applicability,
- object applicability,
- version applicability,
- configuration applicability,
- population applicability,
- environmental applicability,
- jurisdictional applicability,
- operational applicability,
- lifecycle applicability,
- temporal applicability,
- scale applicability,
- domain applicability,
- context applicability,
- evidence transfer,
- applicability equivalence,
- applicability divergence,
- applicability boundaries,
- conditional applicability,
- applicability limitations,
- applicability uncertainty,
- applicability change,
- and applicability traceability.

Module Function

VEAM applies after evidence has been determined sufficiently relevant to one or more Validation Criteria or evidence requirements.

Its function is to prevent:

- evidence from one object being automatically applied to another,
- evidence from an old system version being silently transferred to a new version,
- evidence from one configuration being generalized across materially different configurations,

- population-specific findings being represented as universal,
- evidence from controlled environments being assumed valid in real-world environments,
- jurisdiction-specific evidence being transferred without examining regulatory or contextual differences,
- evidence from normal operating conditions being applied to extreme conditions,
- historical evidence being treated as permanently applicable,
- evidence generated at one scale being generalized to another,
- and similarity being mistaken for methodological equivalence.

Relevance and Applicability Are Different

VALIDOS® establishes a strict distinction:

Evidence Relevance asks:

Does the evidence address the validation question?

Evidence Applicability asks:

Does that evidence address the validation question under conditions sufficiently aligned with the actual validation context?

Evidence may therefore be relevant but inapplicable.

For example: A study examining the same type of AI system may be highly relevant.

If it concerns a materially different model version, population, deployment environment, or operating configuration, its applicability may remain limited.

Applicability Dimensions

VEAM evaluates evidence across the dimensions capable of materially affecting transfer into the governed validation context.

These may include:

Object

Version

Configuration

Population

Environment

Jurisdiction

Operational State

Lifecycle Stage

Time

Scale

Domain

Context

Not every validation requires every dimension.

The applicable dimensions depend upon the Validation Object and Validation Scope.

Object Applicability

Evidence must possess a legitimate relationship to the actual Validation Object.

Evidence generated from:

- the same object,
- an equivalent object,
- a predecessor object,
- a related object,
- a representative object,
- or another object class

may possess different degrees of applicability.

Object similarity alone does not establish applicability.

Material differences must be examined.

Version Applicability

Validation evidence frequently becomes version-sensitive.

A software patch, model update, dataset change, component replacement, algorithm modification, or system redesign may alter whether previous evidence remains applicable.

VEAM therefore asks:

Does the evidence still describe the version being validated?

Where the answer is uncertain, evidence applicability must not be assumed.

Configuration Applicability

The same Validation Object may operate under materially different configurations.

Evidence generated under one configuration may not support another when differences affect the characteristic being validated.

Relevant configuration differences may include:

- model parameters,
- hardware,
- software dependencies,
- safety settings,
- system permissions,
- operational constraints,
- infrastructure,
- or integration architecture.

Applicability must remain configuration-aware.

Population Applicability

Where evidence concerns people, users, patients, customers, operators, datasets, or other populations, VEAM evaluates whether the represented population corresponds sufficiently to the population within the Validation Scope.

Relevant differences may include:

- age,
- language,
- geography,
- behavior,
- demographic composition,
- usage pattern,
- clinical characteristics,
- technical capability,
- or another validation-relevant characteristic.

Evidence from one population should not automatically support conclusions about materially different populations.

Environmental Applicability

Evidence may depend upon the environment in which it was produced.

Relevant environmental dimensions may include:

- physical environment,
- digital environment,
- network conditions,
- infrastructure,

- weather,
- operating load,
- surrounding systems,
- user behavior,
- security conditions,
- or interaction context.

Controlled-environment evidence may remain valuable.

However, its applicability to operational reality must be explicitly governed.

Jurisdictional Applicability

Some evidence depends upon jurisdictional conditions.

These may include:

- laws,
- regulatory requirements,
- institutional practices,
- technical standards,
- healthcare systems,
- financial structures,
- documentation requirements,
- or governance expectations.

Evidence valid within one jurisdiction may remain technically relevant while being insufficient for validation under another jurisdictional environment.

Operational Applicability

Evidence must correspond to the operational conditions represented by the Validation Scope.

Evidence generated during:

Normal Operation

may not automatically apply to:

Degraded Operation

Emergency Operation

Adversarial Conditions

or:

Maximum Load

VEAM prevents one operational state from silently representing all operational states.

Lifecycle Applicability

Validation Objects may behave differently across lifecycle stages.

Evidence from:

- development,
- testing,
- pre-deployment,
- deployment,
- mature operation,
- degraded operation,
- maintenance,
- or retirement

may have different applicability.

Lifecycle transfer requires governance where stage differences are material.

Temporal Applicability

Evidence exists in time.

Its applicability may change because:

- the Validation Object evolves,
- populations change,
- environments change,
- dependencies change,
- operating conditions drift,
- external knowledge improves,
- or previously unknown failure modes emerge.

VEAM therefore evaluates whether historical evidence remains applicable to the current validation.

Scale Applicability

Evidence produced at one scale may not remain valid at another.

Examples include:

- prototype → production system,
 - small dataset → population-scale deployment,
 - laboratory network → national infrastructure,
-

- one autonomous agent → multi-agent ecosystem,
- low transaction volume → high-volume financial operation.

Scale may introduce new dependencies, interactions, failure modes, or emergent behavior.

Evidence transfer across scale therefore requires explicit assessment.

Domain Applicability

A methodology or evidence basis may originate in one domain and be proposed for another.

For example:

Evidence derived from industrial automation may inform autonomous-system validation.

However, domain differences may affect:

- operating assumptions,
- risk conditions,
- user interaction,
- regulatory expectations,
- or failure consequences.

VEAM governs whether cross-domain evidence transfer remains legitimate.

Applicability Equivalence

Evidence transfer becomes stronger where the original evidence conditions and target validation conditions can be shown to be materially equivalent for the validation question.

Equivalence does not require every characteristic to be identical.

It requires that material differences do not invalidate the evidentiary relationship.

VEAM therefore distinguishes:

Identical Conditions

from:

Materially Equivalent Conditions

and:

Materially Different Conditions

Applicability Divergence

An **Applicability Divergence** exists where the evidence-generation conditions and target validation conditions differ in a way that may materially affect evidentiary meaning.

Divergence does not automatically make evidence unusable.

It creates a governance requirement to determine:

- whether transfer remains legitimate,
- whether applicability should be conditional,
- whether additional evidence is required,
- or whether the evidence must be excluded from the target validation role.

Evidence Transfer

VEAM governs the transfer of evidence across conditions.

The governed relationship is:

Source Evidence Context → Material Difference Assessment → Transfer Basis → Target Validation Context

Transfer should possess an identifiable methodological basis.

Statements such as:

"The systems are similar."

or:

"This worked before."

do not by themselves establish evidence applicability.

Transfer Conditions

Evidence transfer may depend upon conditions such as:

- verified equivalence,
- bounded configuration differences,
- unchanged relevant components,
- comparable populations,
- stable operating conditions,
- supplementary evidence,
- independent confirmation,
- or restricted Validation Scope.

Transfer conditions must remain attached to the evidence after applicability determination.

Applicability Boundary

Every evidence item or evidence body may possess a boundary beyond which its applicability cannot be assumed.

An Applicability Boundary may define:

- versions,
- configurations,
- populations,
- environments,
- time periods,
- operational states,
- jurisdictions,
- scale,
- or other conditions.

The boundary prevents validation conclusions from silently expanding beyond the evidence base.

Conditional Applicability

Evidence may be classified as applicable only under explicit conditions.

Examples include:

Applicable only to software version 4.2.

Applicable only to represented patient populations.

Applicable only under nominal network conditions.

Applicable only within the validated operating range.

Applicable only when combined with current operational evidence.

Conditional applicability must remain visible throughout subsequent VALIDOS® governance.

Applicability Uncertainty

Where evidence transfer cannot be confidently established or rejected, VEAM preserves uncertainty.

Evidence may receive:

Applicability Undetermined

rather than being silently accepted or excluded.

This is particularly important where:

- systems are novel,
- evidence is scarce,
- conditions are partially comparable,
- or material differences are poorly understood.

Applicability Status

Relevant Validation Evidence may receive an applicability status such as:

Applicable The evidence can legitimately support the defined validation role under the target conditions.

Conditionally Applicable

The evidence may support the validation role only within explicit conditions or limitations.

Applicability Undetermined

Available information is insufficient to establish legitimate applicability.

Not Applicable

The evidence cannot legitimately be transferred to the target validation context for the proposed role.

These statuses remain specific to the evidence-to-context relationship.

Applicability Is Relationship-Specific

VALIDOS® establishes:

Evidence applicability is not a permanent property of evidence.

The same evidence may be:

Applicable to Object Version A,

Conditionally Applicable to Version B,

and:

Not Applicable to Version C.

Applicability must therefore remain attached to the exact conditions for which it has been assessed.

Applicability and Evidence Gaps

When relevant evidence is found to be inapplicable, an Evidence Gap may emerge.

This is important because evidence may appear to exist until applicability is examined.

For example:

A system may possess extensive historical testing.

If the system has undergone a material architectural change, much of that evidence may no longer apply.

VEAM therefore helps distinguish:

Apparent Evidence Coverage

from:

Applicable Evidence Coverage

Applicability Change Governance

Evidence applicability may change because of:

- object modification,
- version change,
- configuration change,
- population change,
- environmental change,
- jurisdictional change,
- lifecycle progression,
- operational drift,
- scale expansion,
- changed Validation Scope,
- or new knowledge concerning material differences.

Material changes require proportionate applicability reassessment.

Minimum Implementation Framework

1. Establish Evidence Origin Conditions

Identify the material conditions under which the evidence was generated, observed, measured, or recorded.

2. Establish Target Validation Conditions

Identify the corresponding conditions of the actual Validation Object and Validation Scope.

3. Compare Material Dimensions

Assess relevant differences across:

- object,
- version,
- configuration,
- population,
- environment,
- jurisdiction,
- operation,
- lifecycle,
- time,
- scale,
- domain,
- and context.

4. Identify Applicability Divergence

Determine whether any material difference may affect the evidentiary relationship.

5. Establish Transfer Basis

Where evidence is transferred, document the methodological basis supporting that transfer.

6. Establish Applicability Boundary

Define the conditions within which evidence may legitimately support validation.

7. Establish Applicability Status

Classify the evidence as:

- Applicable,
- Conditionally Applicable,
- Applicability Undetermined,
- or Not Applicable.

8. Preserve Applicability Traceability

Maintain:

- Evidence Identifier,
- originating evidence conditions,

- target validation conditions,
- material-difference assessment,
- transfer basis,
- applicability dimensions,
- divergence record,
- applicability boundary,
- conditions,
- limitations,
- status,
- version,
- reassessment triggers,
- and sufficient lifecycle traceability.

Applicability Output

VEAM may produce:

- Evidence Applicability Record,
- Evidence Context Profile,
- Evidence-to-Target Comparison,
- Applicability Divergence Record,
- Evidence Transfer Assessment,
- Evidence Transfer Basis,
- Applicability Boundary,
- Conditional Applicability Record,
- Applicability Determination,
- Applicability Change Record,
- and Applicable Validation Evidence Set.

These outputs support subsequent Evidence Fitness governance.

Use Case 1 — Updated AI Model

Scenario

An AI model has previously undergone extensive validation.

A new version changes the model architecture and training data.

Application

The historical evidence remains relevant to the general validation questions.

VEAM evaluates whether it remains applicable to the new model version.

Evidence relating to unchanged external interfaces may remain applicable, while performance and behavioral evidence affected by the architecture and training changes may require new validation.

Result

Historical evidence is not discarded wholesale, but neither is it automatically transferred.

Applicability is governed at the level of material change.

Use Case 2 — Healthcare Deployment

Scenario

A medical AI system was validated using evidence from hospitals in one country and is proposed for deployment in another.

Application

VEAM assesses differences in:

- patient population,
- clinical workflow,
- language,
- healthcare infrastructure,
- operating conditions,
- and regulatory context.

Result

Some evidence may remain Applicable, some may become Conditionally Applicable, and additional local evidence may be required where material Applicability Divergence exists.

Architectural Position

Validation Evidence Applicability is the second internal governance space of the **Validation Evidence Fitness Standard (VEFS)**.

The progression now becomes:

Candidate Evidence → Evidence Relevance → Evidence Applicability

VERM establishes whether evidence addresses the validation question.

VEAM establishes whether that relevant evidence can legitimately be applied to the actual validation conditions.

Subsequent Evidence Fitness governance can then determine whether the applicable evidence basis provides sufficient coverage, completeness, representativeness, timeliness, and overall fitness.

Foundational Principle

Relevant evidence cannot legitimately support a validation conclusion beyond the conditions to which that evidence applies.

Canonical Closing Statement

Validation Evidence Applicability Module establishes the governance framework through which relevant evidence is tested against the actual conditions of validation before it may support a validation conclusion. By governing object, version, configuration, population, environment, jurisdiction, operation, lifecycle, temporal, scale, domain, and contextual applicability, together with evidence transfer, divergence, boundaries, conditionality, uncertainty, change, and traceability, VEAM prevents validation evidence from being silently generalized beyond the conditions it can legitimately represent.

Parent Resources

→ [VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

→ [VALIDOS® Architecture Map](#)

→ [VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

→ [Previous module: Validation Evidence Relevance Module \(VERM\)](#)

→ [Next module: Validation Evidence Coverage Module \(VECM\)](#)

Related Documents

→ [Validation Evidence Fitness Standard](#)

→ [About Validation Evidence Fitness Standard](#)

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